



Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	An unconfigured edge firewall allowed a malicious actor to execute a DoS attack via an ICMP ping flood. The attack completely overwhelmed the internal network for 2 hours, preventing normal user traffic from accessing corporate resources until incoming ICMP packets were blocked and the network services were restored.
Identify	- Affected assets: Internal network infrastructure, employee workstation, web design services, and critical business application. - Vulnerability identified: An unconfigured edge firewall lacking ICMP inspection, absence of packet rate limiting, and lack of source IP verification against spoofing. - Threat Actor Behavior: Volumetric flooding of ICMP echo requests aimed at consuming total bandwidth and processing capacity to induce network failure.
Protect	- Firewall hardening: Implemented new firewall rules to strictly rate limit incoming ICMP packets, preventing server exhaustion. - Anti spoofing: Configured source IP address verification on the edge firewall to detect and immediately drop spoofed incoming packets. - Baseline configuration: Hardened edge network devices with standardized security configurations to eliminate vulnerabilities. Deploy an IDS/IPS system to

	filter malicious network traffic.
Detect	- Network monitoring software: Deployed continuous traffic analysis tools to monitor bandwidth utilization and automatically flag anomalous spikes in network flow. - IDS/IPS Deployment: Integrated Intrusion Detection and Prevention Systems (IDS/IPS) configured to inspect packet headers and filter out suspicious or malformed ICMP traffic patterns before they penetrate the internal subnet.
Respond	- Immediate Containment: The incident management team blocked all incoming ICMP traffic at the perimeter firewall to stop the flood. - Resource Prioritization: Took non-critical network services offline to conserve memory and bandwidth, ensuring critical business services could be recovered first. - Forensic Triage: Conducted an internal investigation to trace the attack vector back to the unconfigured firewall and implemented long-term defense rules.
Recover	- Service Restoration: Progressively brought critical business services back online under monitored conditions, followed by reactivating secondary non-critical services. - Validation & Testing: Verified that internal staff could access corporate network resources normally without latency or connection drops.

Reflections/Notes:

- Root Takeaway: Availability is just as critical as confidentiality; leaving a perimeter firewall unconfigured leaves the entire enterprise vulnerable to elementary volumetric DoS attacks.
- Proactive Posture: Relying on manual mitigation during an active 2-hour outage creates severe operational downtime. Security defenses must be proactive through automated rate-limiting,

continuous monitoring, and inline IPS inspection.